

Chapter 13: Geometry of Vectors

Recall: For two vectors $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$ in $\mathbb{R}^n$, the **dot product** is

$$\mathbf{x} \cdot \mathbf{y} = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n$$

Definition: Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ be a vector in $\mathbb{R}^n$. Then the **length**, or **magnitude**, or **norm** of the vector $\mathbf{x}$ is

$$\|\mathbf{x}\| = \sqrt{\mathbf{x} \cdot \mathbf{x}}$$

In particular, for $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ in $\mathbb{R}^2$, the magnitude of $\mathbf{v}$ is

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2}$$

Example: What is the length/magnitude of $\mathbf{v} = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$?

$$\|\mathbf{v}\| = \sqrt{(-2)^2 + 5^2} = \sqrt{4 + 25} = \sqrt{29}.$$

Example: What is the length/magnitude of the vector $\mathbf{v} = \begin{bmatrix} 1 \\ 3 \\ -4 \end{bmatrix}$?

$$\|\mathbf{v}\| = \sqrt{1^2 + 3^2 + (-4)^2} = \sqrt{1 + 9 + 16} = \sqrt{26}.$$

Example: Compute the distance between the vectors

$$\mathbf{u} = \begin{bmatrix} 1 \\ -2 \end{bmatrix} \text{ and } \mathbf{v} = \begin{bmatrix} -3 \\ 0 \end{bmatrix}.$$

$$\|\mathbf{u} - \mathbf{v}\| = \left\| \begin{bmatrix} 4 \\ -2 \end{bmatrix} \right\| = \sqrt{4^2 + (-2)^2} = \sqrt{20}$$

Fact: In general, the distance between two vectors $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$ in $\mathbb{R}^n$

is $\|\mathbf{v} - \mathbf{u}\|$, which can be written out as

$$\|\mathbf{v} - \mathbf{u}\| = \left\| \begin{bmatrix} u_1 - v_1 \\ \vdots \\ u_n - v_n \end{bmatrix} \right\| = \sqrt{(u_1 - v_1)^2 + \cdots + (u_n - v_n)^2}$$

Theorem: Let $\mathbf{x}$, $\mathbf{y}$, and $\mathbf{z}$ be vectors in $\mathbb{R}^n$ and let a be a scalar. Then the following properties for dot product hold:

1. $\mathbf{x} \cdot \mathbf{y} = \mathbf{y} \cdot \mathbf{x}$
2. $\mathbf{x} \cdot (\mathbf{y} + \mathbf{z}) = \mathbf{x} \cdot \mathbf{y} + \mathbf{x} \cdot \mathbf{z}$
3. $(a\mathbf{x}) \cdot \mathbf{y} = a(\mathbf{x} \cdot \mathbf{y}) = \mathbf{x} \cdot (a\mathbf{y})$
4. $\|\mathbf{x}\|^2 = \mathbf{x} \cdot \mathbf{x}$
5. $\mathbf{x} \cdot \mathbf{x} \geq 0$ and $\mathbf{x} \cdot \mathbf{x} = 0$ if and only if $\mathbf{x} = \mathbf{0}$
6. $\|a\mathbf{x}\| = |a| \|\mathbf{x}\|$

Proof of _____ *(Pick one):*

$$1. \vec{x} \cdot \vec{y} = x_1 y_1 + \cdots + x_n y_n = y_1 x_1 + \cdots + y_n x_n = \vec{y} \cdot \vec{x}.$$

$$4. \|\vec{x}\|^2 = \left(\sqrt{\vec{x} \cdot \vec{x}} \right)^2 = \left(\sqrt{x_1^2 + \cdots + x_n^2} \right)^2 = \vec{x} \cdot \vec{x}$$

(more space for proof):

$$5. \vec{x} \cdot \vec{x} = \|\vec{x}\|^2 \geq 0, \quad \|\vec{x}\|^2 = 0 \iff \vec{x} = \vec{0}$$

$$6. \|\alpha \vec{x}\| = \sqrt{\alpha \vec{x} \cdot \alpha \vec{x}} = \sqrt{\alpha^2 \vec{x} \cdot \vec{x}} = |\alpha| \cdot \|\vec{x}\|.$$

Theorem: (Cauchy–Bunyakovsky¹–Schwarz inequality)

For every pair of vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$, the following inequality holds:

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|.$$

Proof: We'll do this proof later in more generality (like Bunyakovsky and Schwarz).

If $\mathbf{u}, \mathbf{v}$ are nonzero vectors, we can rewrite the CBS inequality as:

This allows us to define the following:

Definition: The **angle** θ between two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ obeys the formula

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

Example: Let $\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$. What is the angle θ between $\mathbf{u}$ and $\mathbf{v}$?

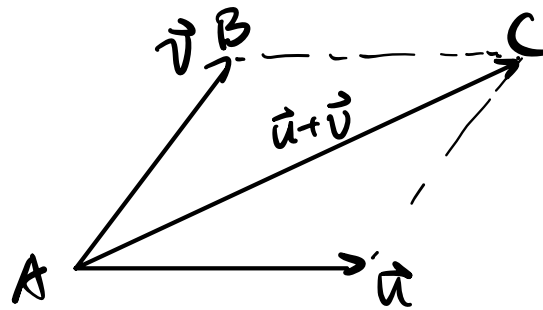
$$\mathbf{u} \cdot \mathbf{v} = 1, \quad \|\mathbf{u}\| = \sqrt{2}, \quad \|\mathbf{v}\| = \sqrt{2}.$$

$$\cos \theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}.$$

¹Most textbooks (including yours) call this the Cauchy-Schwarz inequality. Cauchy proved the most basic version of this theorem, while Bunyakovsky and Schwarz independently proved more general versions of this theorem.

Theorem: (Triangle Inequality) For every pair of vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$, we have $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.

Let's draw a picture to see why this makes sense in $\mathbb{R}^2$:



Look at $\triangle ABC$,
the length of one side,
which is $\|\vec{u} + \vec{v}\|$ is smaller
than that of the other
two.

Proof: (If time):

Question: When will $\|\mathbf{u} + \mathbf{v}\| = \|\mathbf{u}\| + \|\mathbf{v}\|$?

$\vec{u}$ and $\vec{v}$ have the same direction, or
 $\theta = 0$.

Definition: A **unit vector** in $\mathbb{R}^n$ is a vector whose length/magnitude is one.

Fact: If $\mathbf{x}$ is any nonzero vector, then the vector

$$\mathbf{u} = \frac{\mathbf{x}}{\|\mathbf{x}\|}$$

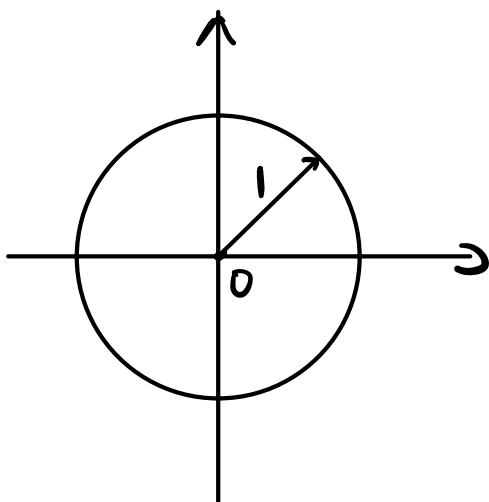
is a unit vector in the direction of $\mathbf{x}$.

Example: Let $\mathbf{x} = \begin{bmatrix} 4 \\ 1 \\ 6 \end{bmatrix}$. Find a unit vector $\mathbf{u}$ in the direction of $\mathbf{x}$.

$$\|\mathbf{x}\| = \sqrt{4^2 + 1^2 + 6^2} = \sqrt{53}$$

$$\vec{u} = \frac{\mathbf{x}}{\|\mathbf{x}\|} = \frac{1}{\sqrt{53}} \begin{bmatrix} 4 \\ 1 \\ 6 \end{bmatrix}$$

Example: Consider the set of all unit vectors in $\mathbb{R}^2$. Visually, what do these create?



A circle of radius 1.